

ANALYSIS OF MACROECONOMIC FACTORS ON STOCK RETURNS

1. DESCRIPTION OF THE DATASET

The dataset consists of monthly data from January 2010 to August 2025 (187 observations) for the following variables:

- HDFC_Stock_Close — dependent variable
- CPI (Inflation)
- IIP (Industrial Production)
- M3 (Money Supply)
- Gold_Price
- Oil_Price
- Exchange_Rate (INR/USD)

Nature of data:

- Secondary sources
- Time-series format
- Monthly frequency
- No missing values

2. RESEARCH PROBLEMS (RESEARCH QUESTIONS)

RQ1: Do macroeconomic variables influence HDFC Bank stock price?

RQ2: Which macroeconomic variable has the strongest effect?

RQ3: Do macroeconomic factors predict movements in HDFC stock returns?

RQ4: Are relationships stable enough for VAR modeling?

3. HYPOTHESES

Primary Hypotheses:

H_0 : Macroeconomic variables have no significant effect on HDFC stock price

H_1 : At least one macroeconomic variable significantly affects HDFC stock price

Granger Causality:

H_0 : Macroeconomic variables do not Granger-cause HDFC stock returns

H_1 : At least one Granger-causes HDFC stock returns

4. DATA-ANALYSIS PLAN

1. Data Preparation: Handle non-stationarity via returns transformation
2. Stationarity Testing: ADF tests on levels and returns
3. Exploratory Analysis: Correlation matrix, time-series plots
4. Econometric Modeling: OLS regression, VIF, Granger causality, VAR, FEVD
5. Interpretation: Connect statistical results to economic meaning

5. EXPLORATORY DATA ANALYSIS

Key Findings:

- Strong upward trends in all variables created spurious correlations (0.97+)
- Conversion to monthly returns resolved stationarity issues
- Gold price showed highest correlation with stock returns (0.17)
- Low pairwise correlations among macro variables in returns format

6. COMPREHENSIVE DATA ANALYSIS

6.1 Stationarity (ADF Tests)

All variables were non-stationary in levels but stationary after returns transformation.

6.2 OLS Regression Results

- $R^2 = 0.047$ (only 4.7% of variation explained)
- Only Gold_Price_Returns significant ($p = 0.034$)
- Coefficient: 0.069 (1% gold increase $\rightarrow$ 0.069% stock increase)

6.3 Multicollinearity Check

VIF values all $< 1.1 \rightarrow$ No multicollinearity in returns data

6.4 Granger Causality Tests

All p-values $> 0.05 \rightarrow$ No macroeconomic variable predicts HDFC stock returns

6.5 VAR Model Stability

VAR models unstable (roots > 1) despite optimal lag selection

6.6 Variance Decomposition

- 80% of HDFC return variation explained by own past values
- Gold prices explain 6%, other macros explain 2-4% each

7. INTERPRETATION OF RESULTS

1. Limited Macro Influence: Macroeconomic variables explain very little of HDFC stock movements (4.7-20%)

2. No Predictive Power: Past macro values don't help predict future stock returns

3. Gold Shows Weak Relationship: Only consistent finding across methodologies, but economically small

4. Market Efficiency: Results support efficient market hypothesis for HDFC

5. Company-Specific Dominance: 80% of movements driven by firm-specific factors

8. CONCLUSIONS & RECOMMENDATIONS

Conclusions:

- HDFC stock price is largely independent of macroeconomic indicators
- Gold price has weak but statistically significant relationship
- Market efficiently incorporates macro information
- Small sample size limits robust econometric modeling

Recommendations:

For Investors:

- Focus on company-specific analysis rather than macro timing
- Use macroeconomic context, not trading signals
- Maintain diversification given high idiosyncratic risk

For Researchers:

- Explore higher-frequency data for short-term relationships
- Investigate nonlinear models or regime-switching approaches

9. LIMITATIONS

1. Small Sample Size: 187 monthly observations limits statistical power

- 2. Monthly Frequency:** May miss higher-frequency relationships
- 3. Model Instability:** VAR models unreliable due to weak relationships
- 4. Linear Assumptions:** May miss complex nonlinear dynamics
- 5. Indian Context:** Findings may not generalize to other markets